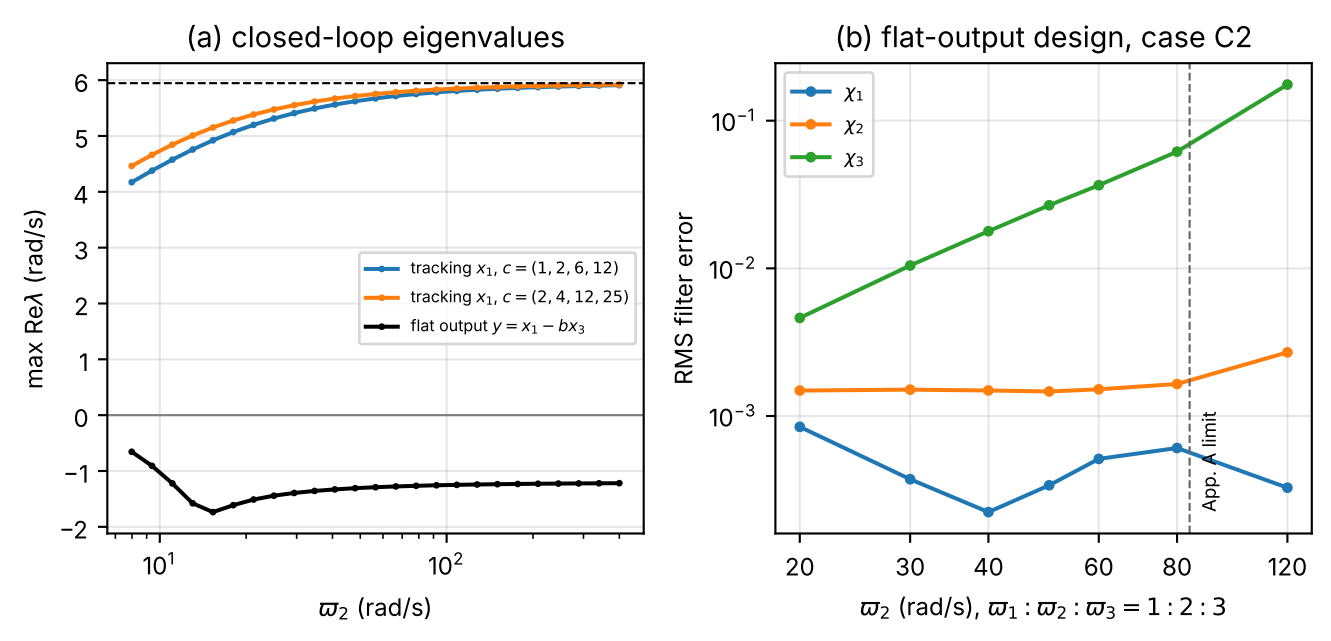
(a) closed-loop eigenvalues
max Reλ (rad/s)
6
5
4
3
2
1
0
−1
−2
10^1
10^2
ϖ_2 (rad/s)
tracking x_1, $c = (1, 2, 6, 12)$
tracking x_1, $c = (2, 4, 12, 25)$
flat output $y = x_1 - bx_3$
(b) flat-output design, case C2
RMS filter error
10^{-1}
10^{-2}
10^{-3}
χ_1
χ_2
χ_3
App. A limit
20
30
40
60
80
120
ϖ_2 (rad/s), $\varpi_1 : \varpi_2 : \varpi_3 = 1 : 2 : 3$